📘 Daily Diary – 24/03/26

🧠 Topic Learned: Hierarchical Clustering

Today I studied **Hierarchical Clustering**, an unsupervised learning method used to group similar data points into clusters.

Hierarchical clustering is often best understood by visualizing how data points "find" each other. Here is a breakdown of the core concepts with visual descriptions.

🌳 The Two Fundamental Approaches

The direction in which you build clusters determines the logic of the algorithm.

- **Agglomerative (Bottom-Up):** Think of this as a "merging" process. Every data point starts as its own tiny island, and the most similar islands join together until only one big continent remains.
- **Divisive (Top-Down):** This is a "splitting" process. You start with one giant group and recursively fracture it into smaller, more specific groups.

🔗 Understanding Linkage Methods

Linkage determines the "closeness" rule. Changing the linkage method can completely change the shape of your clusters.

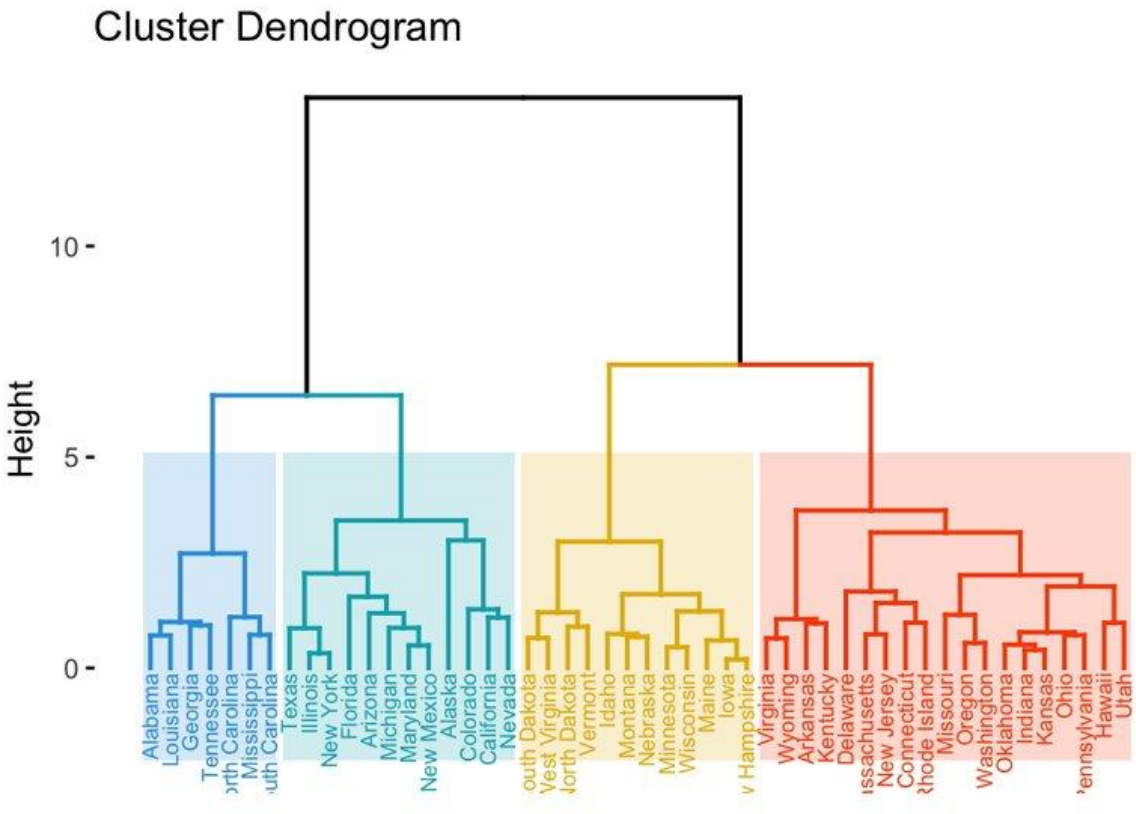
Method	Description	Visual Logic
Single Linkage	Shortest distance between the closest members of two clusters.	Good for capturing long, thin shapes (streaks).
Complete Linkage	Longest distance between the furthest members.	Tends to create compact, spherical clusters.
Average Linkage	The mean of all pairwise distances between the two clusters.	A balanced middle ground; less sensitive to outliers.

Method	Description	Visual Logic
Centroid Linkage	The distance between the geometric center (mean) of Cluster A and Cluster B.	Uses the "heart" of each cluster to measure distance.

📊 The Dendrogram: Your Clustering Map

A **Dendrogram** is a tree-like diagram that records every merge (or split) made during the process.

- **The Y-axis** represents the distance or dissimilarity between clusters.
- **The X-axis** represents individual data points.
- **The "Cut":** By drawing a horizontal line across the dendrogram, you decide how many clusters you want. If you cut higher up, you get fewer, larger clusters; if you cut lower, you get many smaller clusters.



💡 **Real-Life Example: Document Organization**

Imagine you have 100 random news articles.

1. **Bottom-Up:** You group two articles about "Bitcoin" together. Then you group two about "Stock Markets." Eventually, you merge those into a "Finance" cluster.
2. **Top-Down:** You start with "News," split it into "Politics" vs. "Sports," then split "Sports" into "Football" vs. "Cricket."